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Prevalence and correlates of anxiety and depression among ever-married reproductive-aged women in Bangladesh: national-level insights from the 2022 Bangladesh Demographic and Health Survey

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Abstract

Background The sound mental health of reproductive-aged women is crucial for overall maternal and child well-being. However, mental health aspects are historically overlooked in low- and lower-middle-income countries, including Bangladesh. National-level evidence on common mental disorders like anxiety and depression among reproductive-aged women is limited in Bangladesh. Therefore, this study aims to estimate the prevalence of anxiety and depression among ever-married reproductive-aged women in Bangladesh and examine their individual, marital, household, and contextual correlates.

Methods We used data from the Bangladesh Demographic and Health Survey 2022 that collected anxiety and depression data using GAD-7 and PHQ-9 modules, respectively. This study included a total of 19,987 ever-married women aged 15 to 49 years. We used a cluster-adjusted multivariable logistic regression model to examine the factors associated with anxiety and depression.

Results The national level prevalence of anxiety and depression among ever-married reproductive-aged women was 19.5% and 4.9%, respectively. The odds of having anxiety monotonically increased from the age of 25. Menopause, educational attainment, autonomy, household wealth, and type of residence were not associated with anxiety or depression. Non-Muslim women were respectively 34% (AOR: 0.66, 95% CI: 0.55, 0.80) and 33% (AOR: 0.67, 95% CI: 0.97, 2.77) less likely to confront anxiety and depression than Muslims. Having a husband who completed secondary level education, having weekly marital coitus, and residing under the headship of a father or mother-in-law was associated with lower odds of anxiety and depression. Women from the Rangpur and Khulna divisions had higher odds of anxiety and depression than those from Dhaka.

Conclusions This study reveals a high prevalence of anxiety among ever-married reproductive-aged women and highlights that anxiety and depression are not clustered among disadvantageous groups like less educated,

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less autonomous, rural, and poor women. Anxiety and depression are rather associated with late reproductive age, religious affiliation, marital factors, and region.

Keywords Anxiety, Depression, Prevalence, Women, Religion, Pregnancy, Post-partum, Bangladesh

Introduction

Mental health is globally recognized as a universal human right, whereas common mental disorders (CMDs) are often considered a "hidden epidemic" of the twenty-first century [1]. CMDs are characterized by disturbances in mood, cognition, and behavior, which impact an individual's physical, psychological, and social well-being [2]. Having a direct detrimental impact on individuals, CMDs have far-reaching consequences on families, society, and the entire nation [3, 4]. Anxiety and depression are the most prevalent CMDs [2]. Globally, an estimated over 301 million people suffer from anxiety and 280 million from depression as of 2019 [5, 6]. Women face nearly twice the risk of CMDs compared to men [7–9]. An estimated one out of five women faces CMDs, globally [10]. Mental disorders accounted for around 11.47% of Disability-Adjusted Life Years among reproductive-aged (15 to 49 years) women (RAW) [11].

The adverse consequences of CMDs are particularly concerning for RAW, as they usually navigate inevitable gender, social, and economic roles [9]. Maternal depression and anxiety during pregnancy, postpartum, and afterward have been associated with adverse perinatal outcomes, slower fetal development, low birth weight, preterm birth, and being small for gestational age, which often extend into adulthood [12–15]. Infants born to mothers with prenatal depression are at a higher risk of being more irritable, less active, and experiencing developmental delays [12, 16]. Additionally, prenatal depression has been linked to maternal complications such as anemia, diabetes, hypertension, and an increased likelihood of postpartum depression [12, 17].

The onset of CMDs among RAW can be attributable to unique physiological events such as pregnancy, postpartum, and menopause [9]. The prevalence of antenatal depression ranged from 15% to 65% [18]. Another comprehensive review of 565 studies across 80 countries estimated that around 17% of women suffer from postpartum depression [19]. Additionally, RAW experiencing infertility, and transitioning through menopause have a significantly higher prevalence of depression and other mental health disorders [9]. In addition to the unique physiological characteristics of RAW, a range of sociocultural factors can also affect their mental health status. Due to the vast societal differences between low- and lower-middle-income countries (LLMICs) and high-income countries, the burden and reasons for CMDs may vary. In this regard, the social determinants of poor mental health have a higher prevalence in LLMICs [20, 21]. RAW from LLMICs undergo several social deprivations, such as limited access to education, poor social status, childhood abuse, child marriage, low decision-making autonomy, intimate partner violence, and sexual and domestic violence, which are the key drivers of CMDs [22, 23].

Thus, we conceptualized the social framework of CMDs among RAW from LLMICs into four broad domains- individual factors, factors around marriage, household, and contextual factors. At the individual level, a woman's age, education, and religious beliefs shape her ability to cope with psychological distress, while biological changes like pregnancy, postpartum, and menopause can heighten her vulnerability to CMDs [24, 25]. Marital relationships can play a crucial role, as emotional support from a husband may enhance psychological resilience, whereas adverse experiences such as intimate partner violence, widowhood, or separation may significantly increase mental health risks [26–28]. At the household level, living arrangements and family dynamics may also influence psychological well-being. For instance, female-headed households may provide greater autonomy to RAW, extended families can offer social support yet impose additional caregiving responsibilities and limit personal space. Economic stability is another key household-level factor that meets fundamental necessities resulting in psychological stability.

Bangladesh is a South Asian LLMIC with the eighth largest population, globally. Bangladesh is one of the top ten countries with the highest child marriage rate in the world (50.1%) [29, 30], where marriage is usually chosen by the bride's parents or relatives [31, 32]. In Bangladesh, victimization via various forms of violence is common. Victimization through physical violence by a husband or others is historically present in Bangladesh, and sexual harassment and cyberbullying are emerging concerns [31, 33–35]. All these social contexts may make the RAW vulnerable to psychological distress. According to the Bangladesh National Mental Health Survey 2019, 18.8% of adult women had mental health disorders, including depression, anxiety, and stress [36]. India and Nepal, two neighboring countries, have reported a lower prevalence of CMDs than Bangladesh [37, 38].

Mental health of RAW has historically received little attention in Bangladesh, due to the heavy burden of

maternal and child morbidity and mortality [39]. Thus, national-level mental health data on them is also scarce in Bangladesh, resulting in small-scale or region-specific studies limiting generalizability. An earlier systematic review of different hospitals and some community-based studies reported high levels of CMDs among women [40]. An online-based cross-sectional survey among only 451 female university students in Bangladesh found the prevalence of anxiety and depressive symptoms at 69.2% and 45.2%, respectively [41]. An earlier study conducted on 2599 rural RAW estimated the depression prevalence of 20% [42]. Another study in the urban community of Dhaka found a higher risk of CMDs among women [43]. The National Mental Health Survey 2019, also estimated an overall higher prevalence of mental disorders in women than in men (21.5% vs 15.7%) [44]. However, no national survey has been conducted specifically to assess the prevalence and social determinants of anxiety and depression among RAW in Bangladesh.

Knowing national and sub-national level estimates and correlates is crucial for policy formulation and designing interventions to address mental health challenges aimed at improving the overall well-being of RAW in Bangladesh [45]. For the first time, 2022 BDHS collected anxiety and depression data from ever-married RAW, representative at the national, divisional, and rural–urban levels. Hence, using 2022 BDHS data, this study may fill up the existing literature gap by investigating the national-level burden and correlates of anxiety and depression among ever-married RAW. Therefore, we aim to estimate the prevalence of anxiety and depression among ever-married reproductive-aged women in Bangladesh and examine their individual, marital, household, and contextual correlates.

Method

Design, data, and participants

This is an observational study using the most recent publicly available secondary data from the nationwide cross-sectional Bangladesh Demographic and Health Survey (BDHS) 2022 [46, 47]. Bangladesh contains eight administrative divisions: Dhaka, Barishal, Chattogram, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet. Each division is then divided into zilas, and zilas into upazilas.

BDHS 2022 is the 9th round of the DHS survey in Bangladesh that used a two-stage stratified cluster sampling design to make the data representative at the national, divisional, and rural–urban levels. For selecting clusters, the complete list of enumeration areas (clusters) covering the whole country was used. It was prepared by the Bangladesh Bureau of Statistics for the 2011 population census of the People's Republic of Bangladesh. From this list, a total of 674 clusters were selected at the first stage with probability proportional to cluster size, comprising 237 in urban areas and 438 in rural areas. For the second-stage selection of households, a complete household listing operation was then carried out in all selected clusters to provide a sampling frame. Then using this sampling frame, a systematic sample of 45 households per cluster was selected to provide statistically reliable estimates of key demographic and health variables for urban and rural areas separately and for each of the eight divisions in Bangladesh.

The sampling procedure yields a sample of 30,330 households, of which 30,018 were successfully interviewed, resulting in a near 100% response rate. Two-thirds of the households (30 of the 45) in each cluster were randomly selected for the long individual questionnaire that included the mental health section. Thus, a total of 19,987 (unweighted) ever-married RAW received the mental health section. We included all of these 19,987 ever-married RAW in the analysis.

Outcome measure

BDHS 2022 measured anxiety and depressive symptoms using the widely recognized screening tools, the Generalized Anxiety Disorder (GAD-7) and the Patient Health Questionnaire (PHQ-9) [22]. Both scales assessed symptoms reported by participants over the two weeks before the survey.

Anxiety

The GAD-7 assessed anxiety using the following 7 items: (a) feeling nervous, anxious, or on edge; (b) not being able to stop or control worrying; (c) worrying too much about different things; (d) trouble relaxing; (e) being so restless that it is hard to sit still; (f) becoming easily annoyed or irritable; and (g) feeling afraid as if something awful might happen [48]. Each item was scored on a Likert scale ranging from 0–3, respectively, for choices "Never," "Rarely," "Often," and "Always;". So, the total score ranges from 0 to 21. A GAD-7 score of 0–5 is considered mild, while a score of 6–14 is considered moderate and 15–21 is considered severe [48]. A cut-off that distinguishes moderate to severe symptoms from the rest is chosen for the binary categorization of anxiety.

Thus, we constructed the binary outcome of interest "presence of generalized anxiety disorder" as follows:

a. Respondents with GAD-7 scores 6 or higher were considered to have anxiety
b. Respondents who scored less than 6 were considered not to have anxiety

Depression

The PHQ-9 assessed depression using the following 9 symptoms: (a) little interest or pleasure in doing things; (b) feeling down, depressed, or hopeless; (c) trouble falling asleep, staying asleep, or sleeping too much; (d) feeling tired or having little energy; (e) poor appetite or overeating; (f) feeling bad about themselves, or that they are a failure or have let themselves or their family down; (g) trouble concentrating on things such as reading the newspaper or watching television; (h) moving or speaking so slowly that others could have noticed, or the opposite (being so fidgety or restless that they have been moving around a lot more than usual); and (i) thoughts that they would be better off dead or of hurting themselves in some way [49]. Each item was scored on a Likert scale from 0 ("never") to 3 ("always"), with a total score ranging from 0 to 27. A PHQ-9 score of 0–4 is considered minimal symptoms or no symptoms, a score of 5–9 is considered mild, 10–14 is considered moderate, 15–19 is considered moderately severe, and 20–27 is considered severe [50]. A cut-off that distinguishes moderate to severe symptoms from the rest is chosen for the binary categorization of depression.

Thus, we constructed the binary outcome of interest "presence of depression" as follows:

a. Respondents with PHQ-9 scores 10 or higher were considered to have depression
b. Respondents who scored less than 10 were considered not to have depression

Conceptual framework for correlates selection

Problem-solving skills and coping mechanisms help ease psychological upsets [9, 22]. Individual sociodemographic characteristics like age, education, and media exposure help enhance these skills. Access to entertainment and personal autonomy may serve as protective factors against CMDs by promoting relaxation and reducing stress [51, 52]. In addition, high parity reduced personal time, and increased caregiving demands, which can be physically and emotionally draining, contributing to CMDs [53].

Biological stressors-induced hormonal fluctuations may influence CMDs as well. For instance, hormonal fluctuations during different reproductive phases across life-course such as pregnancy, menopause, and postpartum may significantly influence psychological well-being [24, 25].

Marriage is one of the sweetest and most complex institutions that may influence the psychological well-being of a married couple. Emotional and physical connectedness with a partner may foster mental stability [26, 27]. An educated husband may foster economic stability, enhance emotional support, promote shared responsibilities, and reduce domestic violence that leads to a sound psychological state [28, 54]. In contrast, the absence of spousal support can exacerbate financial, social, and emotional burdens which may affect psychological well-being [55]. Thus, several marital factors like marital status, residing with the husband, husband's education, and marital coitus can be the marital correlates of anxiety and depression.

The psychological well-being of an individual largely depends on his/her surroundings. Thus, household-level factors may become influential. For instance, a female household head can be more empathetic to a woman compared to a male head, which may help cope with hardships. Extended families can provide a support network that may buffer against psychological disruption. Moreover, the financial security of the household ensures the fulfillment of basic needs (e.g., food, shelter, healthcare), reducing stress and fostering psychological well-being. Thus, anxiety and depression prevalence can vary by several household-level factors like the gender of the household head, relationship with the household head, household size, and wealth status.

One's psychological well-being is incomplete without community components. Gender roles, availability, and accessibility of women's needs in a region can directly impact psychological well-being. Geographical stressors like extreme heat, air, and water pollution, flood, or drought can also affect mental health. Residence type and divisions are two key correlates that can explain these contextual phenomena. Hence, a differential in anxiety and depression prevalence can be observed across residence types and divisions.

The above discussion summarizes that psychological well-being can be influenced by a complex interplay of factors operating at individual, biological, marital, household, and contextual levels. Thus, we group the covariates into these five broad domains.

Correlates considered

Individual level characteristics, we included women's age (in 5 years age group), educational attainment (no or primary: if woman never attended school or ever attended between class 1 to 5, secondary incomplete: if woman had ever attended between class 6 to 9, secondary complete or higher: if woman had at least 10 years of schooling), television watching (not at all, less than once a week, at least once a week), decision-making autonomy (yes: if a woman makes decisions about own healthcare, major household purchases, and visits to family or relatives, no: otherwise), number of living children (none,1–2, 3+),

religion (Muslim, non-Muslim). Here we considered religious affiliation, because this may shape several daily life factors, gender roles, and household customs that have the potential to influence mental health outcomes.

Biological stressors include menopause status (yes, no), currently pregnant (no or unsure, yes), and currently in post-partum (no, yes).

The marital factors encompassed four variables- current marital status (married, widowed/ divorced/ separated), husband's educational attainment (no or primary: if husband never attended school or ever attended between class 1 to 5, secondary incomplete: if husband had ever attended between class 6 to 9, secondary complete or higher: if husband had at least 10 years of schooling, unknown: husband's education is unknown), currently residing with husband (living with her, staying elsewhere within Bangladesh, staying elsewhere outside Bangladesh), and weekly marital coitus (yes, no).

Household level factors include sex of household head, relationship to household head, household size, and household wealth quintile. The contextual factors included the type of residence and administrative division.

Statistical analysis

There were no missing values in any of the variables except for husband's education. BDHS collects husband's educational information only from currently married women. Thus, the husband's educational information was not asked to 1000 widowed, divorced, or separated women. In addition, 33 currently married women could not report their husband's education. As husband's education was not our outcome of interest, rather than performing missing value imputation, we categorized them into a group, "husband's education unknown". We performed univariate analyses to get a better insight into the data. We estimated the reliability coefficient Cronbach's α to assess the reliability of GAD-7 and PHQ-9 scales using study data [56]. We estimated the prevalence of anxiety and depression across different characteristics of women. The Chi-square test of association was employed to identify the appropriate covariates for regression modeling. Factors that showed an association with anxiety and depression at a 5% level of significance in the Chi-square test were considered for the multivariable regression model. BDHS 2022 used a cluster sampling design, where women are nested within clusters. Due to within-cluster homogeneity in customs and beliefs, anxiety and depression within a cluster can be correlated, which needs to be considered while modeling anxiety and depression. Therefore, we used a survey- and cluster-adjusted multivariable logistic regression model to examine the association of the covariates with anxiety and depression.

From each division, BDHS 2022 sampled the number of women required to get reliable estimates for that division. With this distribution of interviews, some divisions are overrepresented and some divisions are underrepresented compared to the true contribution of each division to the national population share. Therefore, DHS statisticians mathematically calculate a selection probability "weight" that is used to adjust the number of women from each division so that each division's contribution to the total is proportional to the actual population of the division. BDHS 2022 data provides this calculated sampling weight for individual women that adjusts for the unequal selection probabilities. More details about the survey weight calculation can be obtained from the "Sample Probabilities and Sampling Weights" section of the final report [46]. We incorporated survey weight and survey design characteristics of the BDHS 2022 to reduce the bias from the estimates and produce robust standard error of the estimates. All the analyses were done using Stata version 14.0 (Stata SE 14, Stata Corp, College Station, TX, USA).

Results

Sample characteristics

Table 1 presents the distribution of the demographic and socioeconomic characteristics of the analytic sample (reproductive-aged women). Only 25% of the women and their husbands completed secondary or higher education. Nearly half of the women (48.1%) watched television at least once a week and 56% had decision making autonomy in all three domains (own healthcare, major household purchases, and visits to family or relatives). Most of the women were currently married (95.2%) and living with their husbands (79.9%). Among the sampled women, 5.9% were pregnant and 10.5% were in post-partum. The majority of the women were Muslim (90.4%) and came from male-headed households (85%). A substantial portion (71.5%) of the women were from rural areas.

Prevalence of anxiety

The value of the reliability coefficient Cronbach's α for the overall GAD-7 was 0.83, which is greater than the recommended values of 0.70 to 0.80, indicating excellent reliability [56]. About 19.5% of the sample had anxiety. The prevalence of anxiety monotonically increased with women's age (Table 2). Anxiety prevalence was comparatively lower among higher-educated women (14.1%). Husbands' education also had a significant positive influence on women's anxiety. Currently, married women (18.6%) tended to have a lower chance of anxiety than others. On the contrary, women with 3 + children ever born had the highest prevalence of anxiety (24.4%). Muslim women were more anxiety-prone (19.9%) and women

Table 1 Distribution of analytical sample

Factors	Percentage	Weighted Number	Number of Women
Age (in years)			
15–19	8.6	1,729	1,635
20–24	16.4	3,289	3,233
25–29	17.6	3,523	3,539
30–34	17.2	3,437	3,438
35–39	16.7	3,344	3,362
40–44	12.7	2,546	2,561
45–49	10.8	2,160	2,219
Educational attainment			
No or primary	40.2	8,053	7,997
Secondary incomplete	35.6	7,131	6,866
Secondary complete or higher	24.2	4,844	5,124
Television watching			
Not at all	44.4	8,895	8,812
Less than once a week	7.4	1,490	1,482
At least once a week	48.1	9,643	9,693
Decision making autonomy			
No	43.4	8,690	8,882
Yes	56.6	11,339	11,105
Number of living children			
None	10.8	2,169	2,169
1–2	58.6	11,739	11,747
3 +	30.6	6,121	6,071
Religious affiliation			
Muslim	90.4	18,107	17,928
Non-Muslim	9.6	1,921	2,059
Currently pregnant			
No or unsure	94.1	18,848	18,810
Yes	5.9	1,181	1,177
Currently in post-partum period			
No	89.5	17,927	17,923
Yes	10.5	2,101	2,064
Have menopause			
No	96.5	19,328	19,250
Yes	3.5	700	737
Current marital status			
Married	95.2	19,060	18,987
Widowed/ divorced/ separated	4.8	969	1,000
Husband's educational attainment			
No or primary	47.9	9,591	10,430
Secondary incomplete	21.8	4,360	4,258
Secondary complete or higher	25.3	5,071	5,266
Unknown	5.0	1006	1033
Currently residing with husband			
Living with her	79.9	15,999	16,101
Staying elsewhere (within Bangladesh)	6.8	1,361	1,392
Staying elsewhere (outside Bangladesh)	8.5	1,699	1,494
Widowed/ divorced/ separated	4.8	969	1,000

Table 1 (continued)

Factors	Percentage	Weighted Number	Number of Women
Weekly marital coitus			
No	47.1	9,429	9,344
Yes	52.9	10,600	10,643
Sex of household head			
Male	85.0	17,018	17,093
Female	15.0	3,011	2,894
Relationship to household head			
Wife	61.6	12,331	12,342
Daughter	9.1	1,831	1,830
Daughter-in-law	13.8	2,771	2,713
Self	10.3	2,060	1,965
Others	5.2	1,036	1,137
Household size			
1–4	48.2	9,661	9,457
5 or above	51.8	10,367	10,530
Household wealth quintile			
Lowest two	38.0	7,610	7,502
Middle	20.6	4,135	3,989
Upper two	41.4	8,283	8,496
Type of residence			
Rural	71.5	14,328	12,980
Urban	28.5	5,700	7,007
Division			
Dhaka	25.4	5,080	3,028
Barishal	6.0	1,199	2,117
Chattogram	18.7	3,749	2,983
Khulna	11.9	2,389	2,602
Mymensingh	7.6	1,527	2,156
Rajshahi	13.1	2,625	2,546
Rangpur	11.4	2,291	2,399
Sylhet	5.8	1,169	2,156

who belonged to a male-headed household tended to have lower anxiety (19.1%) than female-headed households (21.7%). The highest prevalence of anxiety was found among the women who belong to lower wealth quintile households (21.9%). Women who came from small families were more prone to anxiety (20.4%). Lastly, rural women had a higher prevalence (20.1%) of anxiety than urban women. Anxiety was independent of women's autonomy in the Chi-Square test.

Correlates of anxiety

Results of multivariable regression (Table 3) suggest that at least one factor from all the domains of our conceptual framework showed an association with anxiety.

Individual-level factors

Women's age, education, religious affiliation, and being in postpartum were individual-level factors associated with anxiety. Women aged 35–49 had 69% more odds of having anxiety compared to women aged 15–24 years (AOR: 1.69; 95% CI: 1.43–1.99). Non-Muslim women had 34% lower odds of having anxiety than Muslim women (AOR: 066; 95% CI: 0.55–0.80).

Marital factors

Husband's educational attainment and weekly marital coitus were associated with lower odds of anxiety. The odds of anxiety were 24% lower among women whose husbands had completed secondary or higher education than women whose husbands had no education (AOR: 0.76; 95% CI: 0.66–0.89).

Table 2 Prevalence of anxiety and depression among ever-married reproductive-aged women in Bangladesh

Factors	Anxiety			Depression		
	Prevalence	95% CI	Chi-Square (*p*-value)	Prevalence	95% CI	Chi-Square (*p*-value)
Age (in years)			34.744 (< 0.001)			8.269 (< 0.001)
15–19	12.9	[10.9, 15.2]		3.7	[2.7, 4.9]	
20–24	12.9	[11.5, 14.4]		3.2	[2.6, 4.0]	
25–29	17.0	[15.4, 18.6]		4.6	[3.9, 5.5]	
30–34	20.8	[19.1, 22.7]		5.2	[4.3, 6.2]	
35–39	23.5	[21.7, 25.3]		5.4	[4.6, 6.4]	
40–44	23.5	[21.5, 25.6]		5.4	[4.4, 6.5]	
45–49	25.9	[23.8, 28.1]		7.3	[6.2, 8.5]	
Educational attainment			67.138 (< 0.001)			9.032 (< 0.001)
No or primary	23.4	[22.0, 24.9]		5.7	[5.0, 6.4]	
Secondary incomplete	18.7	[17.4, 20.0]		4.8	[4.2, 5.5]	
Secondary complete or higher	14.1	[12.9, 15.4]		3.8	[3.1, 4.6]	
Television watching			3.691 (0.026)			0.015 (0.984)
Not at all	20.6	[19.2, 22.1]		4.9	[4.3, 5.7]	
Less than once a week	18.2	[15.9, 20.8]		4.8	[3.7, 6.3]	
At least once a week	18.6	[17.5, 19.9]		4.9	[4.4, 5.6]	
Decision making autonomy			0.061 (0.804)			0.014 (0.906)
No	19.6	[18.3, 20.9]		5.0	[4.4, 5.6]	
Yes	19.4	[18.3, 20.6]		4.9	[4.3, 5.6]	
Number of living children			57.78 (< 0.001)			7.415 (< 0.001)
None	15.9	[14.1, 17.7]		4.8	[3.8, 6.0]	
1–2	17.6	[16.6, 18.6]		4.4	[4.0, 5.0]	
3 +	24.4	[22.8, 26.0]		5.9	[5.2, 6.7]	
Religious affiliation			11.431 (< 0.001)			3.635 (0.057)
Muslim	19.9	[18.9, 21.0]		5.0	[4.5, 5.6]	
Non-Muslim	15.4	[13.3, 17.8]		3.8	[2.8, 5.0]	
Currently pregnant			7.041 (0.008)			0.004 (0.949)
No or unsure	19.7	[18.7, 20.7]		4.9	[4.5, 5.5]	
Yes	16.3	[14.0, 19.0]		4.9	[3.6, 6.5]	
Currently in post-partum period			39.811 (< 0.001)			13.447 (< 0.001)
No	20.2	[19.1, 21.2]		5.1	[4.6, 5.7]	
Yes	13.6	[11.9, 15.6]		3.2	[2.4, 4.1]	
Have menopause			7.6 (0.006)			2.332 (0.127)
No	19.3	[18.3, 20.4]		4.9	[4.4, 5.4]	
Yes	23.9	[20.5, 27.6]		6.2	[4.6, 8.4]	
Current marital status			149.835 (< 0.001)			55.421 (< 0.001)
Married	18.6	[17.6, 19.7]		4.7	[4.2, 5.2]	
Widowed/ divorced/ separated	36.3	[32.9, 39.9]		10.2	[8.4, 12.4]	
Husband's educational attainment			73.451 (< 0.001)			22.772 (< 0.001)
No or primary	21.6	[20.3, 23.0]		5.4	[4.7, 6.1]	
Secondary incomplete	17.5	[16.1, 19.1]		4.4	[3.7, 5.3]	
Secondary complete or higher	14.0	[12.7, 15.3]		3.5	[2.9, 4.2]	
Unknown	35.6	[32.4, 39.1]		10.1	[8.3, 12.2]	

Table 2 (continued)

Factors	Anxiety			Depression		
	Prevalence	95% CI	Chi-Square (*p*-value)	Prevalence	95% CI	Chi-Square (*p*-value)
Currently residing with husband			41.373 (< 0.001)			16.529 (< 0.001)
Living with her	19.0	[18.0, 20.0]		4.8	[4.3, 5.3]	
Staying elsewhere (within Bangladesh)	18.1	[15.7, 20.9]		4.7	[3.6, 6.1]	
Staying elsewhere (outside Bangladesh)	15.6	[13.1, 18.5]		3.7	[2.6, 5.2]	
Widowed/ divorced/ separated	36.3	[32.9, 39.9]		10.2	[8.4, 12.4]	
Weekly marital coitus			28.699 (< 0.001)			31.967 (< 0.001)
No	21.4	[20.1, 22.7]		5.9	[5.3, 6.7]	
Yes	17.8	[16.7, 19.0]		4.0	[3.5, 4.6]	
Sex of household head			7.118 (0.008)			2.055 (0.152)
Male	19.1	[18.1, 20.2]		4.8	[4.3, 5.4]	
Female	21.7	[19.7, 23.8]		5.5	[4.6, 6.5]	
Relationship to the household head			27.615 (< 0.001)			8.173 (< 0.001)
Wife	20.5	[19.4, 21.7]		5.2	[4.7, 5.9]	
Daughter	19.3	[17.1, 21.7]		4.8	[3.8, 6.1]	
Daughter-in-law	11.5	[10.1, 13.0]		2.7	[2.1, 3.5]	
Self	23.6	[21.4, 26.1]		6.0	[5.0, 7.3]	
Others	20.5	[17.7, 23.5]		5.1	[3.8, 6.7]	
Household size			6.551 (0.011)			5.315 (0.021)
1–4	20.4	[19.2, 21.7]		5.4	[4.8, 6.0]	
5 or above	18.6	[17.4, 19.9]		4.5	[4.0, 5.1]	
Household wealth quintile			13.788 (< 0.001)			3.351 (0.035)
Lowest two	21.9	[20.4, 23.5]		5.4	[4.7, 6.2]	
Middle	19.1	[17.5, 20.8]		5.4	[4.4, 6.5]	
Upper two	17.4	[16.2, 18.8]		4.3	[3.7, 5.0]	
Type of residence			3.893 (0.049)			1.782 (0.182)
Rural	20.1	[18.9, 21.3]		5.1	[4.6, 5.8]	
Urban	17.9	[16.3, 19.7]		4.4	[3.6, 5.3]	
Division			5.142 (< 0.001)			4.852 (< 0.001)
Dhaka	16.6	[14.7, 18.7]		4.2	[3.2, 5.3]	
Barishal	18.9	[16.2, 21.8]		4.7	[3.7, 6.1]	
Chattogram	21.4	[18.6, 24.5]		4.2	[3.0, 5.9]	
Khulna	20.1	[17.8, 22.6]		6.5	[5.3, 8.0]	
Mymensingh	17.0	[14.5, 19.9]		3.8	[2.9, 4.9]	
Rajshahi	17.8	[15.3, 20.6]		3.7	[3.0, 4.7]	
Rangpur	25.6	[22.7, 28.8]		7.6	[6.0, 9.6]	
Sylhet	20.3	[16.9, 24.1]		6.5	[4.7, 8.8]	

Household-level factors

From this domain, only the relationship with the household head was associated with anxiety. Women whose father-in-law was the household head had 27% lower odds of having anxiety than the women whose husband was the household head (AOR: 0.73; 95% CI: 0.62–0.87).

Contextual factors

Compared to women from the Dhaka division, the odds of anxiety were respectively 1.69 and 1.25 times among women residing in the Rangpur and Khulna divisions.

Prevalence of depression

The value of the reliability coefficient Cronbach's α for the overall PHQ-9 was 0.83, which is greater than the

Table 3 Correlates of anxiety and depression among ever-married reproductive-aged women in Bangladesh

Factors	Anxiety		Depression	
	AOR	95% CI	AOR	95% CI
Age (in years)				
15–24	Reference		Reference	
25–34	1.45***	[1.27, 1.67]	1.36***	[1.09, 1.69]
35–49	1.69***	[1.43, 1.99]	1.38***	[1.09, 1.75]
Educational attainment				
No or primary	Reference		Reference	
Secondary incomplete	0.99	[0.90, 1.10]	1.09	[0.91, 1.31]
Secondary complete or higher	0.85**	[0.73, 0.99]	1.06	[0.80, 1.40]
Television watching			Not retained	
Not at all	Reference			
Less than once a week	0.98	[0.82, 1.17]		
At least once a week	1.06	[0.96, 1.18]		
Decision making autonomy	Not retained		Not retained	
No				
Yes				
Number of living children				
None	Reference		Reference	
1–2	0.91	[0.78, 1.06]	0.74**	[0.56, 0.98]
3 +	1.04	[0.86, 1.25]	0.84	[0.60, 1.16]
Religious affiliation				
Muslim	Reference		Reference	
Non-Muslim	0.66***	[0.55, 0.80]	0.67**	[0.49, 0.92]
Currently pregnant			Not Retained	
No or unsure	Reference			
Yes	1.03	[0.86, 1.24]		
Currently in post-partum period				
No	Reference		Reference	
Yes	0.83**	[0.70, 0.98]	0.77	[0.58, 1.03]
Have menopause			Not retained	
No	Reference			
Yes	1.10	[0.89, 1.34]		
Current marital status				
Married	Reference		Reference	
Widowed/ divorced/ separated	2.17	[0.81, 5.84]	1.17	[0.29, 4.69]
Husband's educational attainment				
No or primary	Reference		Reference	
Secondary incomplete	0.90	[0.80, 1.01]	0.90	[0.73, 1.11]
Secondary complete or higher	0.76***	[0.66, 0.89]	0.73**	[0.56, 0.94]
Unknown	0.88	[0.34, 2.28]	1.53	[0.40, 5.84]
Currently residing with husband				
Living with her	Reference		Reference	
Staying elsewhere (within Bangladesh)	1.06	[0.85, 1.31]	1.06	[0.78, 1.45]
Staying elsewhere (outside Bangladesh)	0.82	[0.65, 1.04]	0.79	[0.53, 1.19]
Unknown				
Weekly marital coitus				
No	Reference		Reference	
Yes	0.87***	[0.79, 0.96]	0.67***	[0.57, 0.78]

Table 3 (continued)

Factors	Anxiety		Depression	
	AOR	95% CI	AOR	95% CI
Sex of household head				
Male	Reference		Not retained	
Female	1.06	[0.85, 1.32]		
Relationship to household head				
Wife	Reference		Reference	
Daughter	1.10	[0.92, 1.31]	0.83	[0.60, 1.16]
Daughter-in-law	0.73***	[0.62, 0.87]	0.62***	[0.46, 0.84]
Self	0.86	[0.65, 1.14]	0.80	[0.59, 1.09]
Others	0.91	[0.74, 1.13]	0.76	[0.53, 1.09]
Household size				
1–4	Reference		Reference	
5 or above	0.95	[0.86, 1.04]	0.92	[0.78, 1.09]
Household wealth quintile				
Lowest two	Reference		Reference	
Middle	0.91	[0.81, 1.03]	1.10	[0.88, 1.38]
Upper two	0.92	[0.81, 1.04]	0.93	[0.75, 1.17]
Type of residence				
Rural	Reference		Not retained	
Urban	0.98	[0.85, 1.13]		
Division				
Dhaka	Reference		Reference	
Barishal	1.13	[0.89, 1.43]	1.12	[0.76, 1.63]
Chattogram	1.40***	[1.12, 1.74]	1.04	[0.67, 1.61]
Khulna	1.25**	[1.02, 1.54]	1.62***	[1.15, 2.28]
Mymensingh	0.95	[0.74, 1.20]	0.87	[0.60, 1.28]
Rajshahi	1.01	[0.81, 1.27]	0.85	[0.59, 1.23]
Rangpur	1.69***	[1.36, 2.10]	1.88***	[1.30, 2.72]
Sylhet	1.19	[0.92, 1.54]	1.55**	[1.02, 2.35]

** $p < 0.05$, *** $p < 0.01$

recommended values of 0.70 to 0.80, indicating excellent reliability [56]. Table 2 explores the potential factors associated with experiencing depression among reproductive-aged women. About 4.9% of the women had depression. Like anxiety, depression prevalence monotonically increased with age. Women and their husbands' educational attainment had a positive influence on women's depression status. The prevalence of depression was the lowest in currently married women (4.7%) and highest in divorced women (10.2%). The Chi-Square test indicates that women with 3 + children ever born had a significantly higher prevalence of depression (5.9%). The prevalence of depression was significantly higher among women who were from female-headed households (5.5%) than male-headed households (4.8%). Rural women had a higher prevalence of depression than urban (5.1% vs 4.4%). Watching television, women empowerment, and current pregnant status were insignificant in bivariate analysis at a 20% level of significance.

Correlates of depression

Results of multivariable regression (Table 3) suggest that at least one factor from all the domains of our conceptual framework showed an association with depression.

Individual-level factors

Women's age, religion, and number of living children were the individual-level correlates of depression. Women aged 35–49 had 38% higher odds of experiencing depression than women aged 15–24 years (AOR: 1.38; 95% CI: 1.09–1.75). The odds of having depression were 33% lower in non-Muslim women than in Muslims (AOR: 0.67; 95% CI: 0.49–0.92).

Marital factors

Husband's educational attainment and weekly marital coitus were associated with lower odds of depression. Women whose husbands had completed secondary or higher education had 27% lower odds of having depression than women whose husbands had no education (AOR: 0.73; 95% CI: 0.56–0.94).

Household level factors

From this domain, only the relationship with household head was associated with depression. Women residing under the headship of a father or mother-in-law had 38% less odds of confronting depression than women residing under the headship of their husbands (AOR: 0.62; 95% CI: 0.46–0.84).

Contextual factors

Women from Khulna, Rangpur, and Sylhet divisions had significantly higher odds of experiencing depression compared to women from Dhaka, with increases of 62%, 88%, and 55%, respectively.

Discussion

Main findings

This study reveals that one in five ever-married reproductive age women is suffering from anxiety and every 20th had depression. Anxiety and depression were more prevalent among women aged thirty or above. Muslim women were more likely to confront anxiety and depression than non-Muslims. Higher education of husband and weekly marital coitus was associated with a lower chance of anxiety and depression. Women residing under the headship of a father or mother-in-law showed less anxiety and depressive symptoms. Women from the Rangpur and Khulna divisions showed higher odds of anxiety and depression, but no differential was observed across rural and urban areas.

Comparison with earlier studies

The WHO estimates the global prevalence of depression and anxiety among women to be approximately 5.1% and 4.6%, respectively [57]. The burden of CMDs is heavier among Bangladeshi women than women from two neighboring countries, India and Nepal [37, 38]. However, CMD prevalence among Bangladeshi women is somewhat lower than observed in another neighboring country, Pakistan. The correlates of anxiety and depression among RAW we found include age, education, religion, current marital status, husband's education, weekly marital coitus, relationship with household head, and administrative division. Similar to our findings, studies from neighboring countries like India, Nepal, and Pakistan have also demonstrated the association of anxiety and depression with factors such as age, marital status, religion, and geographic regions [58–61].

Individual characteristics and anxiety-depression

Age is a crucial marker of CMDs. Psychological distress tends to peak among middle-aged and older women, driven by a combination of life stressors, including menopausal transitions, health-related challenges, and family responsibilities [62]. We found a significantly higher level of anxiety and depression among women of middle or later reproductive age, which is consistent with earlier studies from neighboring countries [58, 59, 61]. Interestingly, we found no association of experiencing menopause with anxiety and depression. An earlier cohort study published in the "World Psychiatry" revealed that mental health issues between ages 41 and 50 were likely not directly linked to the menopausal transition [63]. They also concluded that earlier reported associations may be false positives resulting from improper dichotomization, reporting bias, undisclosed multiple adjustments, or overfitting.

Education has a guiding effect on mental well-being by enabling better coping mechanisms, employment opportunities, and financial stability [64]. Our findings reveal a significantly lower burden of anxiety among secondary-level educated women that reconfirms the positive role of education in enhancing mental well-being. Studies from India and Nepal reported a similar association between anxiety and educational attainment [59, 65].

Religion can disrupt psychological well-being through the funnel of several societal aspects because religious customs and regulations are often practiced according to the social customs of a region. Inappropriate social adaptation of religious customs often imposes many regulations which may trigger mental health stressors. In addition, distinct gender roles across religious groups may lead to differentials in accessing education, occupation, and practicing autonomy, which are key social determinants of mental health status. Minority induced discrimination and social insecurity may cause psychological stress to the minor religious groups. Therefore, a differential in mental disorders across religious groups is not an uncommon phenomenon. In this study, we found a higher prevalence of anxiety and depression among Muslim women than among non-Muslims.

An earlier study from 1988 showed that the distinct subordinate cultural regulations and social expectations within the patriarchal Muslim community in Bangladesh may impose additional stress on women in certain circumstances [66]. However, this may not explain our findings on CMD and religion. The regression model from which we concluded for religion and anxiety-depression association, controlled for the key determinants of the

social funnel (women's age, media exposure, education; husband's education; household composition, and wealth) by which religion may influence mental well-being. We excluded women's autonomy from that model because this was not even associated in the Chi-square test. This indicates that the religion and anxiety-depression association we observed was due to something other than the social factors we mentioned. Importantly, Muslims are the majority (91%) in Bangladesh, which nullifies the minority issue, as well [67]. Moreover, we found no earlier evidence from any Muslim majority country that explored CMD differential by religious groups. Thus, further dedicated investigation is recommended to explain the religious differentials in mental health outcomes.

Marital factors and anxiety-depression

A harmonious relationship between a married couple is crucial for their psychological well-being. The absence of spousal support often exacerbates financial, social, and emotional burdens affecting the psychological well-being of reproductive-aged women [55]. Earlier studies from neighboring countries like India, Nepal, and Pakistan found a higher risk of CMDs among widowed, divorced, and separated women compared to currently married women [58, 59, 61]. However, we found no differentials in CMDs among widowed, divorced, and separated women compared to currently married women. This suggests that the absence of a husband does not exacerbate financial, social, and emotional burdens among Bangladeshi women to such an extent that can lead to CMDs.

Husband's education contributes to a sound mental state among married women by fostering economic stability, enhancing communication and emotional support, promoting shared responsibilities, and reducing the likelihood of domestic violence [54]. Consistent with findings from neighboring countries, higher levels of husband's education were associated with lower rates of anxiety and depression [68, 69]. In addition, we found lower levels of anxiety and depression among married women engaged in regular coitus. Engaging in regular sexual activity fosters emotional bonding and enhances physical intimacy between partners, which helps reduce stress and improve mental well-being [27].

Role of household and contextual factors on CMDs

The gender of the household head does not have any association with women's anxiety or depressive symptoms. In addition, women residing under the headship of a father or mother-in-law showed significantly lower symptoms of anxiety and depression than women residing under the headship of their husbands. The possible reasons for lower psychological disruption may include a reduced burden of financial responsibilities, household maintenance, and decision-making. However, further research is needed to be conclusive. The number of household members did not make any differentials in CMD prevalence. We found no socioeconomic differentials in adverse mental health outcomes, suggesting programs covering all socioeconomic groups equally. We found a significant regional variation in anxiety and depression. Women from the Rangpur and Khulna divisions had the highest odds of experiencing anxiety and depression. Geographical variation in mental health outcomes was also revealed in Nepal [70].

Adequacy of government actions to improve mental health

Mental health services are inadequate throughout the country and highly centered in the capital and big cities [36]. In recent years, the government of Bangladesh (GoB) has taken several initiatives to address the mental health needs of citizens. Bangladesh is among the first few countries in WHO SEAR to place mental health as one of its top 10 priority health conditions [71]. The GoB enacted the Mental Health Act 2018 and approved the National Mental Health Strategic Plan 2020–2030 for implementation [72]. To accomplish Sustainable Development Goal 3.4, the Ministry of Health and Family Welfare pledged in its fourth five-year Health Population and Nutrition Sector Program 2017–2022 to make "mental health and psychosocial well-being" a priority area within the existing health care system [73]. However, estimates say there is just one psychiatrist for every 200 thousand people and the average amount spent by the government on mental health is only 0.08 USD per person yearly [71].

Recommendations

Contact points for introducing mental health aspects

An inclusive approach is crucial to ensure all women are aware of mental health rights and services. Initiating mental health awareness in secondary grade textbooks may help, as girls' enrollment in secondary grades is rising. Community clinics, maternal and child welfare centers, and upazila health complexes are on the list of primary care-seeking points for women. The availability of booklets describing mental health aspects in these care-seeking points can be helpful. Digital self-screening systems using simple tools like the GAD-7 and PHQ-9 in health facilities can facilitate early identification and care-seeking.

Integrate mental health care in maternity care

The high ANC and PNC coverage offers an opportunity to screen and manage CMDs among pregnant and post-partum women. We recommend integrating CMDs screening into existing maternal and child health

programs like ANC and PNC. In addition, developing national guidelines for prenatal and post-partum CMDs screening and treatment are essential.

Prioritize the high-risk divisions

Given the high risk of anxiety and depression among women in Rangpur and Khulna, targeted mental health interventions should be prioritized in these regions. The National Mental Health Strategic Plan 2020–2030 has emphasized expanding mental health care in primary and secondary level health facilities and the GoB is also testing its feasibility in some districts [72]. Ministry of Women and Child Welfare established 67 One-Stop Crisis Centers (OCC) that provide psychological counseling to female survivors of violence and trauma [72]. We recommend scale-up and strengthening these facilities in Khulna and Rangpur divisions. For better accessibility and quality services, the GoB should expand community-based awareness programs to break the stigma around mental health and strengthen mental health workforce capacity. In addition to these, the GoB can think of digital mental health services to improve accessibility in remote areas of Khulna and Rangpur divisions.

Investment in future research

Investment in mental health research is required to understand the patterns and reasons better, and design culturally acceptable and implementable innovative solutions. Some specific areas of future research are highlighted below:

The low prevalence of depression estimated using the 2022 BDHS is suspicious. Three years back, the Bangladesh Adolescent Health and Wellbeing Survey 2019 reported a depression prevalence of 15.2% among currently married female adolescents (aged 15 to 19), whereas the 2022 BDHS estimated a much lower prevalence of 3.7%. A further investigation comparing these two data sources would help explain this difference. Region-specific research is highly recommended to identify unique cultural or geographical stressors contributing to high prevalence in some regions.

This study finds some interesting results that deviated from the expected direction. For example: higher CMDs among Muslims despite being the majority, widowed/divorced/separated women did not show higher levels of CMDs than the currently married group, post-partum women confront less CMDs than the rest, women under the headship of a father or mother-in-law had low CMD prevalence. Further research in these areas is required to explore the underlying factors.

Strengths and limitations

For the first time, this study estimates the national-level prevalence of anxiety and depression among ever-married RAW in Bangladesh and identifies the correlates using data from the nationwide BDHS 2022. The main strengths of this study include the use of mental health screening tools, GAD-7 and PHQ-9, which have high sensitivity and specificity in detecting anxiety and depression. The sampling strategy of BDHS 2022 enables interpreting results at national, divisional, and rural–urban levels. The high response rate and data completeness of BDHS 2022 strengthen the analyses. We performed the analyses using appropriate survey weight and survey design characteristics that reduce bias from the estimates and yield robust standard errors for the estimates.

Establishing any causal relationship between the factors associated with anxiety and depression is beyond the scope of the study. Due to the secondary data type, we could not adjust for some potential covariates such as previous mental health histories, other psychological stressors, other comorbidities, social connectedness, and experience of any form of violence that may lead to anxiety or depression.

Conclusion

The findings challenge conventional assumptions by showing that anxiety and depression are not clustered among disadvantaged groups. Hence, an inclusive approach to mental health strategies is essential. Community-level mental health interventions should be tailored to the vulnerable groups. For example: the association between marital coitus and a lower prevalence of anxiety and depression suggests that community-level mental health interventions should also focus on strengthening marital relationships. Here, communication and conflict resolution training for the couples might be effective. Prioritizing the inclusion of low-educated men in interventions is crucial, as women with less-educated husbands showed higher mental health disruptions. Local mental health awareness campaigns in high-risk regions are crucial for early diagnosis. In addition, digital screening systems in health facilities can improve early diagnosis. In conclusion, we anticipate that this study lays the groundwork for further research, designing inclusive, culturally sensitive, and regionally adapted mental health policies and interventions.

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Authors' contributions
MMR and MTA conceptualized the study. MTA and TA performed data curation, formal analysis, visualization, and writing of the original draft. SNE and ZF contributed to the literature review. MTA and TA prepared the original draft with the help of BP, ZF, and MHP. MMR critically reviewed and provided constructive supervision. MTA incorporated comments from the authors. All authors reviewed and approved the final draft of the manuscript.

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Data availability
BDHS 2022 data is publicly available on the DHS website.

Declarations

Ethics approval and consent to participate
As this study used two national-level secondary data that are publicly available upon reasonable request, no Ethics Committee or Institutional Review Board approval is required.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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